

Emotion Classification Pipeline

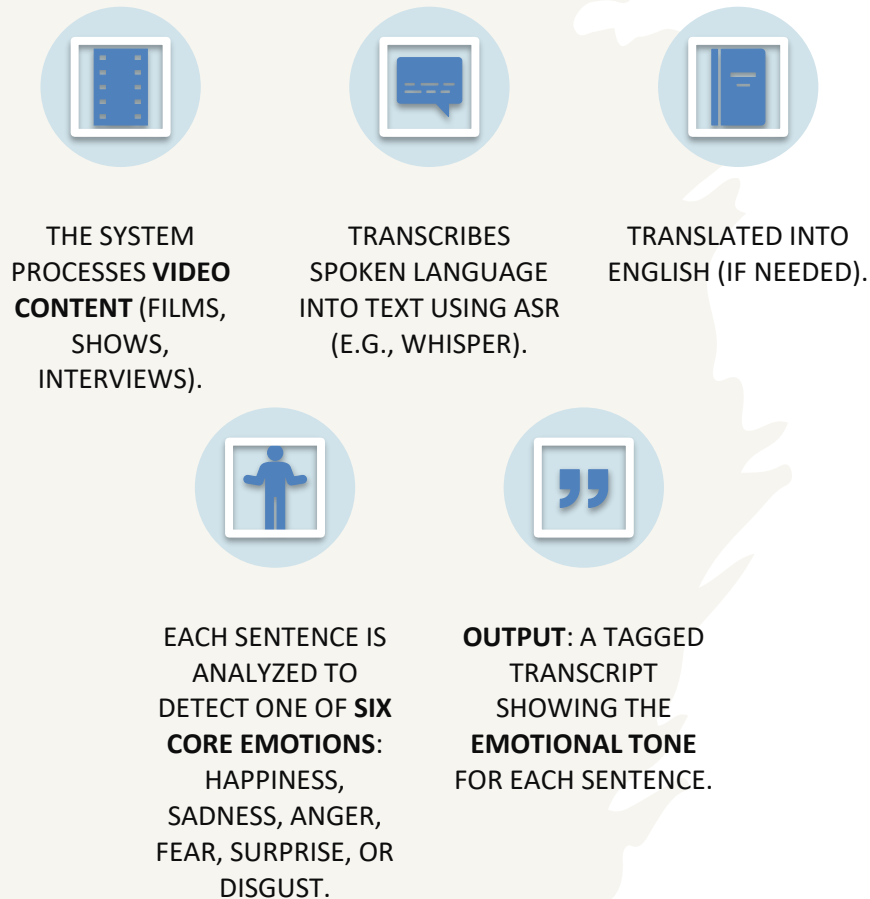
Presentation | Arabic 6 Team

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Use Case – Emotion Detection in Media Content

Real-time Emotion Classification from Media Transcripts



Value for the Client – Content Intelligence Agency

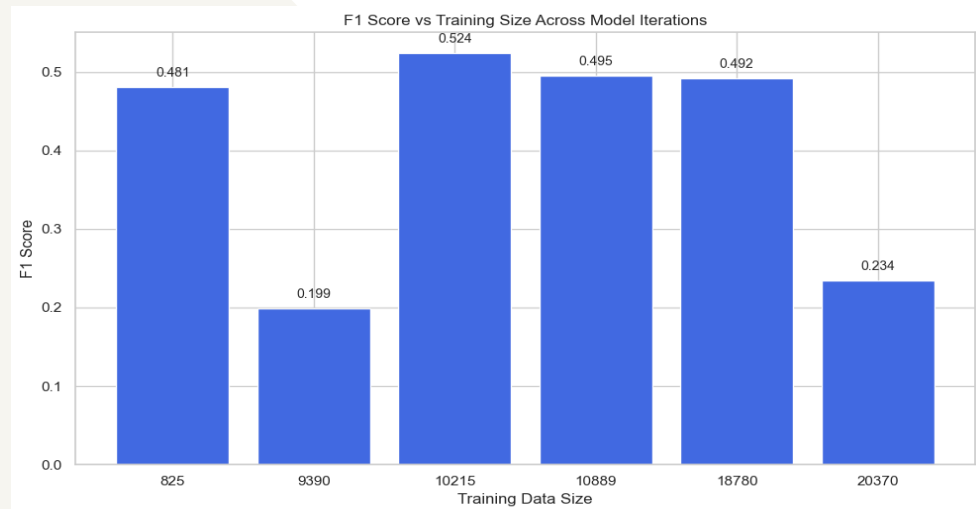
Enhancing Media Analysis through Emotion AI

- Helps **media creators** and editors **understand emotional dynamics** in their content.
- Enables **emotion-based indexing** for better **searchability** and **content recommendations**.
- Supports **audience targeting** and **editorial decisions** by analyzing emotional arcs.
- Can be integrated into the client's existing **LLM-powered pipeline** for automation and scalability.
- Reduces the need for **manual tagging**, improving speed and consistency.

Data Characteristics:

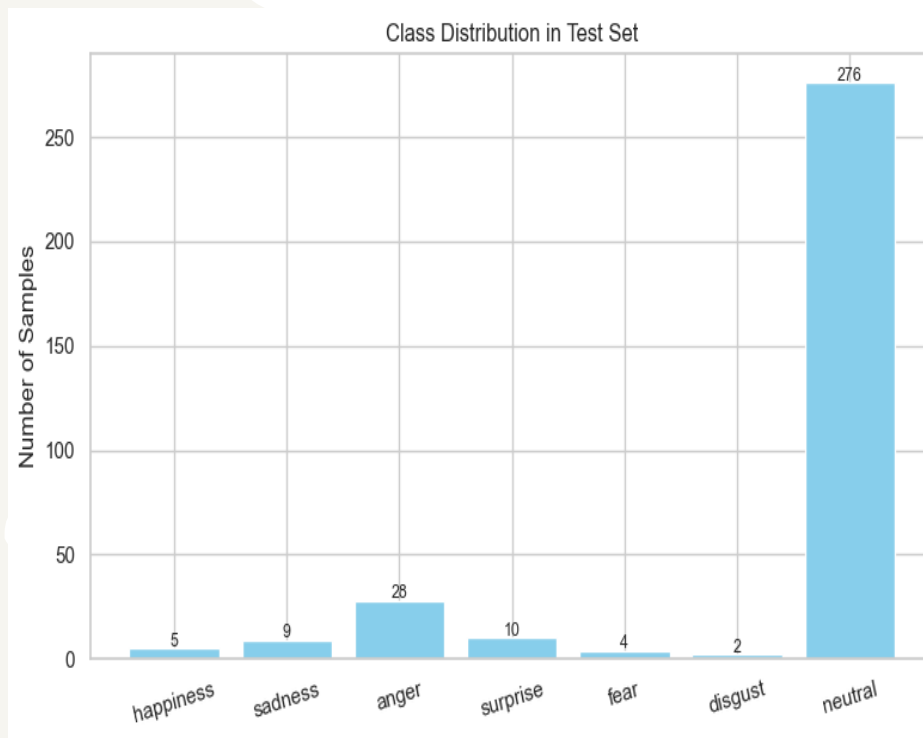
- **Datasets used for Training:**

- Kaggle Dataset (~13,000 tweets)
- Agency Dataset from Another Arabic Group (~825 examples)
- English Dataset (GoEmotions ~70,000 sentences)
- Train on the first video and test on the 2nd



- **Limitations:**

- Imbalanced Emotion Distribution
- Dialect diversity
- Impact of Translation Quality
- Limited Availability of Emotion-Labeled Arabic Datasets



Models

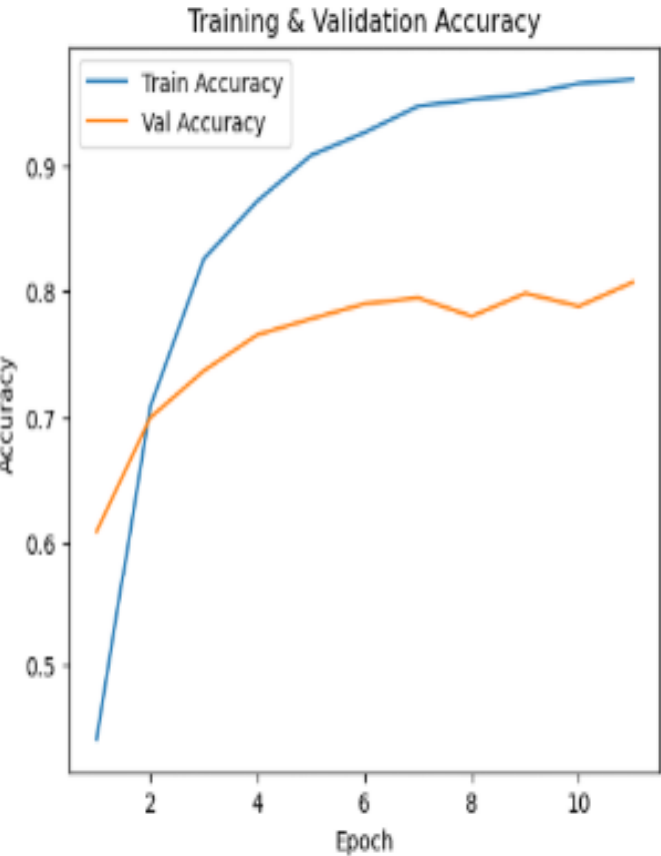
MODEL	TYPE	HOW IT WORKS	BEST FOR	WHY USE IT
Logistic Regression	Linear Model	Model Predicts probabilities based on a linear relationship.	Simple problems, binary classification	Fast, easy to interpret
Naïve ayes	Probabilistic Model	Uses Bayes' theorem to predict class probabilities.	Text classification (e.g., spam detection)	Simple, fast, works well with high-dimensional data
RNN	Deep Learning (Sequential)	Processes sequential data by passing info from one step to the next.	Time-series, text, speech	Handles sequences efficiently, faster to train than LSTM
LSTM	Deep Learning (Sequential)	Processes sequential data by passing info from one step to the next.	Complex sequences (e.g., language tasks)	More complex and slower than RNN, but better for long-range dependencies
Transformer	Deep Learning (Attention-based)	Uses self-attention to process input in parallel, capturing relationships.	NLP tasks (e.g., translation, sentiment analysis)	Fast, great for complex language tasks, handles long-range dependencies

Model Performance & Insights

Metric	Logistic Regression	Naïve_bays	LSTM
Accuracy	0.76	0.80	0.71
Precision	0.14	0.13	0.19
Recall	0.15	0.14	0.20
Weighted F1_score	0.73	0.74	0.72
Mode type	Linear	Probabilistic	Deep Learning

Model Performance & Insights

In this graph the training set and the testing set was different from what the agency classified, which we put to showcase how crucial the dataset was in Terms of getting high accuracy



Metric	RNN	Transformer
Accuracy	0.365	0.82
Precision	0.45	0.17
Recall	0.37	0.17
Weighted F1_score	0.32	0.77
Mode type	Deep Learning (Sequential)	Deep Learning (Attention-based)

Transformer model iterations

Iteration	F1-score	What we changed
First iteration	'eval_f1': 0.494863	for this iteration we tried to change things one at each stop we start by changing the checkpoint model I replaced MODEL_CHECKPOINT = "xlm-roberta-base" Multilingual model with MODEL_CHECKPOINT = "aubmindlab/bert-base-arabertv02"
Second iteration	'eval_f1': 0.5238557	in this iteration it had improved by 2 percent still nothing significant #in the next iteration i will tweak a little bit we start by making the batch size 16 ✅ Reduce Warnings & Improve Label Mapping Instead of <code>train_df["label"] = train_df["Emotion"].map(label_map)</code> , use <code>.apply()</code> to avoid NaN issue Use batch size control and <code>padding="longest"</code> to reduce unnecessary padding
Third iteration	eval_f1': 0.76866	This was our best model, in this iteration we fixed the emotion labeling in the test dataset and rised the number of epochs

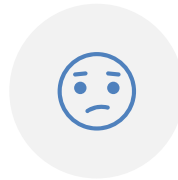
Error Analysis



**IMBALANCED
EMOTION
DISTRIBUTION**



**SENTENCE LENGTH
& CLARITY**



**SIMILAR
EMOTIONS
CONFUSED**



**TRANSLATION
ISSUES**



**CONFUSION
MATRIX INSIGHTS**

Ethical Considerations

- **Potential Dialect Bias:** The model may systematically overlook or misinterpret certain dialect expressions, especially those from underrepresented communities or less-common speech patterns.
- **Subjectivity in Emotion Labeling:** Emotions can overlap or manifest differently across cultures and annotators. Certain filler words or frequent tokens may be misleadingly emphasized as emotional cues.
- **Transparency in Decision-Making:** While we've analyzed attention patterns to understand predictions, some explanations still highlight non-emotive words. This can impact user trust if the model consistently focuses on the wrong tokens.



Limitations Of the pipeline

Manual Translation Bottleneck

- Current pipeline relies on translation steps for handling Arabic content, slowing down data processing and risking inconsistency.

Partial Arabic Coverage

- Model depends on English-focused techniques, which don't fully capture Arabic linguistic nuances or morphological complexity.

Rare Emotions & Class Imbalance

- Some emotions (e.g., disgust, fear) appear too infrequently for robust modeling, making the classifier biased toward more common emotions.

Dialect Variation

- Different Arabic dialects are underrepresented, causing the pipeline to misinterpret dialect-specific vocabulary or grammar.

Next Steps

Adopt

Adopt Direct Arabic Models

- Explore specialized Arabic architectures (e.g., AraBERT) to reduce dependence on manual translation and improve linguistic accuracy.

Refine

Refine Explanations

- Enhance explanation methods to ensure the pipeline focuses on genuinely emotive tokens rather than filler words or frequent tokens.

Expand

Expand Data Collection

- Gather additional examples for underrepresented emotions and integrate multiple dialects to improve coverage and reduce bias.

Optimize

Optimize Resource Usage

- Investigate more efficient model architectures or training procedures to manage computational costs and environmental footprint.

Conclusion

End-to-end Arabic
emotion pipeline
developed



Ready for client use
with ongoing
improvements



Next: deploy +
improve with direct
Arabic modeling

